

Automated UI Test Framework

Tech stack: Java · Selenium · TestNG · Maven · Docker · Jenkins

1. Project Overview

This repository shows a complete CI/CD pipeline for browser-based UI testing:

- Dynamic UI tests driven by TestNG
- Headless Chrome in Docker + Xvfb
- Jenkins Pipeline for build, test, and reporting

2. Key Components

1. **Source & Tests**

- `src/test/java/.../DynamicUITComponentsTest.java`
- Generates its own TestNG XML (`dynamic-suite.xml`)

2. **Maven (`pom.xml`)**

- Declares Selenium, TestNG, and other dependencies
- Installs a custom JAR (`seleniumUpgrade-0.0.1-SNAPSHOT.jar`) into the local

repo

3. **Docker**

- **Dockerfile**

- Base image: `maven:3.8.7-eclipse-temurin-17`
- Installs Google Chrome and Xvfb
- Copies code into `/app`
- Includes `entrypoint.sh` to start Xvfb and run tests

- **entrypoint.sh**

```
```bash
#!/bin/bash
set -e
echo "Starting Xvfb on display :99"
Xvfb :99 -screen 0 1280x1024x24 &
export DISPLAY=:99
echo "Running Maven tests"
mvn clean test -B -Dheadless=true -Dsurefire.suiteXmlFiles=dynamic-
```

suite.xml

```

- **Usage**

```
```bash
Build the Docker image
docker build -t testing-docker:latest .
Run tests inside a container—requires only Docker in your environment
docker run --rm \
 -v "$PWD:/app" \
 -v "$PWD/.m2:/root/.m2" \
 -w /app \
 testing-docker:latest
```
```

3. Intentional Test Failure

One test is designed to fail to demonstrate pipeline reporting:

| Test Method | Expected vs. Actual | Result |
|----------------------------|-----------------------|-------------|
| selectDrillAndSetQuantity` | asserts 4 but finds 1 | FAIL |

Observe:

- Jenkins test reports
- Failure annotations in console and XML

4. Benefits

- **Consistency:** Docker + Xvfb deliver identical environments everywhere.
- **Visibility:** Jenkins captures every build, test, and failure in detail.
- **Scalability:** Easily add tests or browsers without changing CI logic.